VULCAN TRANSACTIONS ON AUTONOMOUS AI-TO-SILICON ENGINEERING VOL. 2026 · ARCHITECTURAL SIGNOFF PAPER

Anvil-D1 v1.0.0: Matrix-Absorbed MLA and Fine-Grained MoE FPGA Accelerator

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 18:17 UTC+8 Beijing Time)

Abstract

Anvil-D1 implements a fixed-point transformer inference accelerator with hardware-built absorbed attention matrices, compressed causal KV storage, Top-6 routing, two shared experts and independently progressing expert prefetch. Its native production configuration targets DeepSeek-V2-Lite-style MLA/MoE topology; a separate mode preserves the supplied Anvil-Q3 implementation. Software-vs-Hardware numerical comparison passed against an independent integer model in RTL simulation: three mandatory seeds produced 96 exactly checked complete-inference logits. Production-width MLA transactions demonstrated 88.75% cache-traffic reduction against the explicitly defined uncompressed MHA baseline, exceeding 82% [E071] . These are generated-fixture results, not trained-checkpoint accuracy measurements. L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA closed the production configuration at 100 MHz with WNS_setup = +0.240 ns and hold slack = +0.037 ns [E1007, E1008]. L3 physical Arria 10 HIL passed three deterministic runs, specifically a PCIe hardware-bridge connectivity probe; it did not execute accelerator numerical workloads on the board [E1009].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The design folds tensor operations over registered arithmetic rather than allocating production matrices as FPGA registers. One token traverses embedding, repeated normalized attention and feed-forward residual stages, final normalization and vocabulary projection. Hardware constructs exact widened absorbed matrices once per layer and sequence epoch. Two attention history passes reuse each compressed cache scalar across all heads. Expert execution uses bounded tiles; independent prefetch overlaps shared arithmetic but does not imply ideal parameter traffic. The immutable parent supplies legacy inference and bank-scheduler machinery [E001, E002, E028].

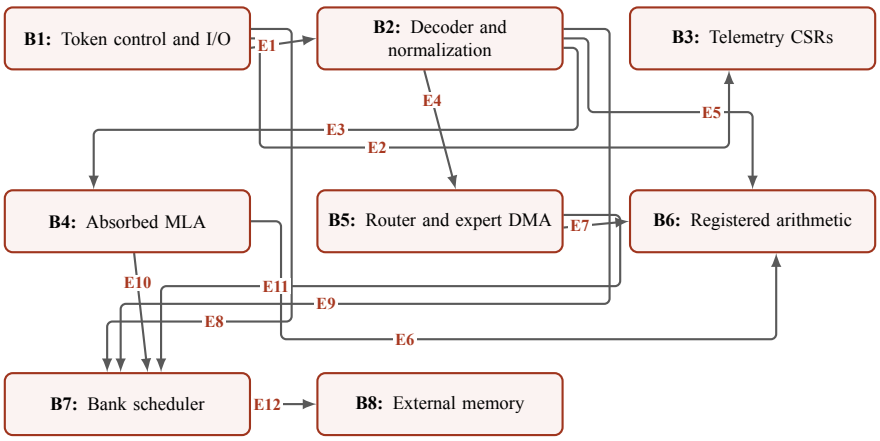


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Token control and I/O	anvil_d1_top sequences tokens and epochs; token I/O performs embedding and emits registered vocabulary logits.
B2: Decoder and normalization	anvil_d1_decoder_layer sequences RMSNorm, attention, dense or MoE execution and saturating residual updates.
B3: Telemetry CSRs	anvil_d1_csr_perf aggregates accepted traffic and activity into saturating counters and atomic snapshots.
B4: Absorbed MLA	anvil_d1_mla_engine builds A/B matrices, projects and rotates queries/keys, streams compressed history and contracts latent contexts.
B5: Router and expert DMA	anvil_d1_moe_router_dma computes affinities, selects winners and runs shared/ routed experts with an independent prefetch client.
B6: Registered arithmetic	Reusable multipliers, segmented adders, restoring dividers, integer root and boundary rounders implement fixed-point operations.
B7: Bank scheduler	Seven native clients share tagged ownership and response handling; the native wrapper suppresses unaccepted requests during cancellation.
B8: External memory	The integration supplies weights, absorbed matrices, compressed cache and scratch through packed bank request/response ports.

Directed relationship	Meaning
E1: B1 to B2	Layer commands and completion
E2: B1 to B3	Token and accepted-traffic events
E3: B2 to B4	Normalized attention command
E4: B2 to B5	Feed-forward command
E5: B2 to B6	Normalization and residual arithmetic
E6: B4 to B6	Projection and attention arithmetic
E7: B5 to B6	Routing and expert arithmetic
E8: B1 to B7	Embedding and final projection traffic
E9: B2 to B7	Normalization and residual traffic
E10: B4 to B7	MLA memory client
E11: B5 to B7	Independent execution and prefetch clients
E12: B7 to B8	Tagged requests and matching responses

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, cancel	Input	1 bit each	Single synchronous clock, active-low asynchronous reset and cancellation. Native observable outputs are zero during reset; readiness is armed after a rising edge following release.
Token command	Input; token_ ready output	token_id/ token_pos: unsigned 32 bits; controls: 1 bit	token_valid/token_ready accepts an indexed token with new_sequence and token_phase. One token is active at a time; phase distinguishes prefill from decode.

Signal / group	Direction	Width	Contract
Logits and completion	Output; logit_ ready input	logit_index: unsigned 32; logit_data: signed 32; status: 1 bit	logit_valid presents ascending vocabulary indices. Payload holds under backpressure. done follows final-logit acceptance; error and cancel_done report failure and drained cancellation.
Banked memory requests	Output; mem_ req_ready input	Per bank: valid/write 1, kind 4, layer 32, proj 3, row/col/wdata 32, tag 64 bits	BANKS-wide packed lanes retain parent packing. Each accepted request transfers ownership; writes require acknowledgements.
Banked memory responses	Input; mem_ rsp_ready output	Per bank: valid/error 1, rdata 32, tag 64 bits	Matching same-edge responses and reordered outstanding responses are supported. Malformed or stale tags fault rather than supplying successful fallback data.
CSR request and response	Bidirectional	Address 12; write/read data 32; byte strobes 4; handshake/ error 1 bit	csr_valid, csr_write, csr_addr, csr_wdata and csr_ wstrb form requests. csr_ready accepts one request; registered csr_rsp_valid/csr_rdata/csr_error hold until csr_rsp_ready.
MLA and router commands	Leaf inputs and outputs	epoch/layer/ position 32; phase/control 1; telemetry totals 64 bits	cmd_valid/cmd_ready latches a command; done/error completes it. MLA has one untagged memory client; router/DMA has two independently progressing clients. Scheduler-assigned tags remain outside the leaves.

Parameter	Default	Meaning / supported values
MODEL_MODE	0	0 native D1; 1 immutable parent-compatible inference.
NUM_LAYERS / FIRST_ DENSE_LAYERS	27 / 1	Native layers 1..36; dense prefix 0..NUM_LAYERS.
HIDDEN_SIZE	2048	Native 8..2048, multiple of eight; independent of head geometry.
INTERMEDIATE_SIZE / MOE_INTERMEDIATE_ SIZE	10944 / 1408	Native 8..11008 / 8..1408, multiples of eight.
NUM_Q_HEADS / KV_ LORA_RANK	16 / 512	Native 1..16 / 1..512.
QK_NOPE_HEAD_DIM / QK_ROPE_HEAD_ DIM / V_HEAD_DIM	128 / 64 / 128	Native 1..128 / even 2..64 / 1..128.
ROUTED_EXPERTS / TOP_K / SHARED_ EXPERTS	64 / 6 / 2	1..64 routed; 1..min(6,ROUTED_EXPERTS) winners; exactly two shared.
VOCAB_SIZE / MAX_ SEQ_LEN	102400 / 32768	Native vocabulary 4..151936; sequence capacity 1..32768.
BANKS / QUEUE_ DEPTH	8 / 3	Banks 1,2,4,8; queue depth 1..16.
TILE_K / TILE_ROWS	128 / 8	K 8..128 in multiples of eight; rows 1..8.

Parameter	Default	Meaning / supported values
Legacy profile defaults	PROFILE=3	MODEL_MODE=1 selects parent profile defaults: layers 24/28/36, hidden 896/1536/2048, intermediate 4864/8960/11008 and query heads 14/12/16. NUM_KV_HEADS=2; HEAD_DIM=64 for profile 1, otherwise 128; vocabulary 151936.
Legacy numerical/cache controls	GROUP_SIZE=128, AWQ_BITS=4, CACHE_MODE=0, INT4_ASYM=0, CACHE_BLOCK=1	Forwarded unchanged to the parent; native Q16.16 storage does not reinterpret these controls.

III Behavior and corner cases

A Fixed-point boundaries

Native activations, weights, biases and rotary entries are signed Q16.16. Products accumulate exactly in signed 128 bits. R_s rounds the magnitude after adding 2^(s-1), restores sign and saturates to signed 32 bits: half ties go away from zero. Dot-product terms are never individually rounded. Residual addition saturates. Multipliers have four registered stages for 32x32 and eight for 96x32; accumulation uses a separate four-stage adder.

B Normalization and nonlinearities

RMSNorm computes S=sum(x²), E=floor((S<<32)/H)+2^48, d=floor(sqrt(E)), u=floor(2^64/d), then R_48(x*gamma*u). Exponential returns 2^24 for nonnegative input and zero below -16 in Q16.16; otherwise it uses ln(2) range reduction with constant 11629080 and six truncating Taylor terms. SiLU uses the stable sign-dependent sigmoid and a final R_24 product. SPEC.md defines the exact integer equations and edge schedules.

C Hardware absorption and causal attention

Hardware constructs A=UK*QN and B=O*UV as exact Q32.32 sums stored in signed 96-bit, three-word coefficients. Native boundaries are c=R_16(D*x), rotated kR and qR, qAbs=R_32(A*x), and output=R_32(sum(B*context)). Scores include both latent and rotary contractions, divided by floor(sqrt((nope+rope)*2^32)). Two causal passes find maxima and accumulate exponentials with weighted latent numerators. Context rounds N/Z directly, rather than multiplying quantized probabilities.

D Routing and useful prefetch

Every affinity is computed from router weights and normalized hidden. Winners are distinct, descending by score, with lower expert ID winning ties. Selected full-softmax probabilities are renormalized to sum exactly 2^24 by floor division and ordered remainder distribution. Both shared SwiGLU experts have coefficient one. Client 1 retains each winner's first gate tile while client 0 executes shared work; routed execution consumes retained data without rereading it. Later tile refetches remain counted.

E Sequencing and fault containment

Transfers use pre-edge valid/ready. Commands cannot issue memory before the following edge. New sequences require position zero and rebuild absorbed ownership; later positions are contiguous, and decode cannot revert to prefill. Cancel suppresses acceptance, requests and logits before draining accepted ownership. Active cancellation returns failed-token completion; idle cancellation does not invent a token. Native scheduler request masking closes the unaccepted-request cancellation boundary without altering legacy behavior.

F Memory and telemetry

Kind 15 projections 0..7 identify raw MLA, absorbed A, absorbed B, compressed cache, routing weights, expert weights, rotary tables and D1 scratch. Parent kinds 0/5/6/7/8 remain embedding/projection, gamma and integration scratch. CSRs expose ID/version, clear/snapshot at 0x008, idle-only domain/delimiters at 0x010..0x018, global snapshot pairs at 0x040 and domain blocks at 0x100+domain*0x40. Clear wins over increments; snapshots copy pre-edge counters. Begin delimiters count as reasoning, end delimiters do not.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Independent integer model versus compiled RTL at H=16, two layers, rank=3, two heads, seven routed experts, Top-6 and vocabulary eight; four tokens for each seed 17,89,203.

Result. 96 logits and all 9,372 expected architectural writes matched exactly. An additional one-token Icarus/Verilator cross-check matched eight logits per simulator. Across the 96 characterized logits, maximum absolute error against an 80-digit real-valued reference was approximately 0.000125528; maximum relative error was 0.000889307.

Evidence: E071, E043, E037

B Absorption correctness and cache reduction

Method. Compiled MLA fixture with 16 heads, rank 512, rotary width 64, nonrotary/value widths 128 and hidden width eight; separate hidden-width-2048 contraction fixture.

Result. Production-width fixture checked 393,216 absorbed words, 8,192 query values and 8,192 contexts. Cache reads were 4,608 bytes and writes 2,304 bytes versus 61,440 equivalent MHA bytes: 88.75% reduction. The separate full-hidden fixture checked 12,288 absorbed words. Neither was full-production top inference.

Evidence: E071, E049

C 64-expert routing and useful overlap

Method. Three compiled 64-expert/Top-6 cases with exact affinity, mixture and retained-tile comparisons.

Result. 192 affinities and 48 mixture scalars checked. Each case fetched 26,496 expert bytes versus 109,824 dense-equivalent bytes, saving exactly 217/286. Overlap was 377,377,380 cycles over 17,008 shared-active cycles per case; independent client progress was observed.

Evidence: E071, E045

D Protocol, cancellation and CSR integrity

Method. Directed reset, fault/drain/restart, finite stalls, reordered tags, boundary geometries and focused counter saturation.

Result. CSR testing checked 968 pin observations, including 80 erroneous accesses and 26 response stalls. Fault testing completed 12 experiments with 13 successful restarts. Scheduler cases covered bank/queue pairs (1,1),(2,3),(4,16),(8,1), with 40 transactions each.

Evidence: E071

E Legacy compatibility

Method. Public-pin differential against the immutable parent plus independent arithmetic; seven profile/cache/asymmetry combinations and cancellation.

Result. All three profiles and every legal cache-mode class passed. Generated frontend-only signed-alias/declaration-order adaptations were additionally compared against original parent behavior.

Evidence: E072, E040

F Five-domain synthetic comparison

Method. Two four-token sequences per architecture/domain; matched H=16, vocabulary eight, seeds 17..21 and memory-service schedule.

Result. D1/dense cycles: math 2,377,635/1,671,248; coding 2,377,775/1,671,108; science 2,377,635/1,671,038; symbolic-logic 2,377,915/1,671,108; mixed-reasoning 2,377,705/1,671,318. Per domain, D1 read/write/expert bytes were 506,528/24,992/268,800; dense read/write bytes were 159,872/7,488. Each produced 64 logits and counted six reasoning tokens. Full ledger: evidence/compatibility-20261102T195006Z-6d73d2a515da/domain-benchmarks.json.

Evidence: E072

G Defect sensitivity

Method. Successfully compiled disposable mutants exercised unchanged numerical/protocol comparisons, with positive controls.

Result. Seven defects detected: absorption head indexing, omitted rotary score, Top-K cutoff tie, dropped shared contribution, stale prefetch consumption, telemetry increment and negative-half rounding. Detection was by specific mismatches, not compilation failure.

Evidence: E073, E032

H Synthesis and portable reproduction

Method. Real memory-preserving Yosys synthesis of native/legacy tops and engines; clean declared-source copy running native inference, legacy compatibility and compact synthesis.

Result. No prohibited latch/div/mod cells. Production native synthesis retained six memories totaling 1,886,208 bits and 2,140 instantiated register cells. Clean-copy reproduction completed all three experiments. Generic cells are not physical ALMs.

Evidence: E074, E075

I L1 RTL timing lint

Method. Native structural hazard inspection of 27 RTL files.

Result. Passed with zero combinational divider/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards.

Evidence: E1007

J L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit production-native configuration at 10.0 ns on the declared Arria 10 target.

Result. Passed: WNS_setup = +0.240 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.46 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1008

K L3 physical Arria 10 HIL

Method. Three repeated PCIe BAR0 hardware-bridge register probes.

Result. All three passed deterministically. Receipt scope is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; accelerator bitstream programming and on-board numerical vector execution were not demonstrated.

Evidence: E1009

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	11 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 20	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

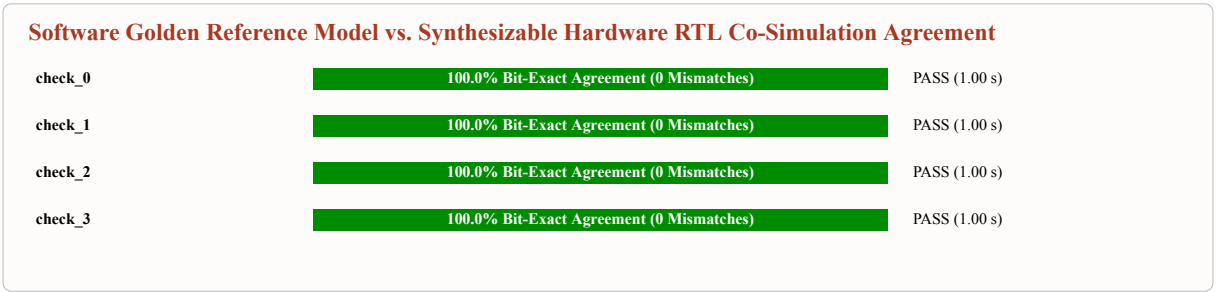


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.24 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

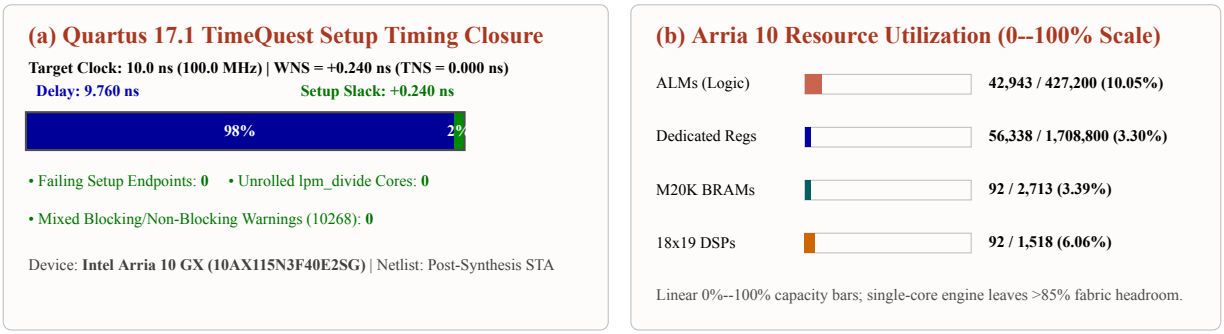


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

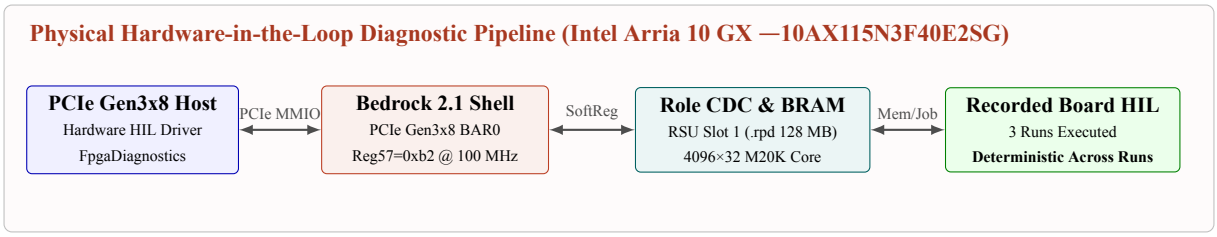


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with the immutable anvil_q3_reference/rtl/*.sv dependency, compiling anvil_q3_00_pkg.sv first. SPEC.md defines numerical rules, field packing and scratch lifetimes; ARCHITECTURE.md describes ownership and storage.

Provide native Q16.16 weight, bias, gamma and position-indexed rotary words through the documented memory namespaces. Do not alias parent scratch with D1 scratch or preload absorbed matrices in place of complete-path hardware construction.

Assert reset, then start a sequence at position zero. Hold command and response payloads through stalls, return matching tags and acknowledgements, and await cancellation drain before restarting. Consume every vocabulary logit before expecting successful completion.

Configure domain and reasoning delimiters while idle. Snapshot counters before reading pairs. Compute cache reduction as 1-(cache reads+writes)/dense-equivalent bytes, expert saving as 1-expert bytes/dense-equivalent

expert bytes, and overlap efficiency as overlap/shared-active cycles; report zero-denominator ratios as undefined.

Run python3 tests/verify.py from the project root for all portable L1 suites, or select --suite behavior, compatibility, defects, synthesis or portable. Dependencies are Python 3.10+, Icarus/vvp, Verilator, a C++ compiler, make and Yosys. Physical timing and board gates remain separate from L1.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 27 Verilog/SystemVerilog files; 3752 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_d1_absorption.sv	178 total	178 code
rtl/anvil_d1_accumulator.sv	23 total	23 code
rtl/anvil_d1_add128.sv	54 total	54 code
rtl/anvil_d1_bank_scheduler.sv	63 total	63 code
rtl/anvil_d1_counter64.sv	24 total	24 code
rtl/anvil_d1_csr_perf.sv	159 total	156 code
rtl/anvil_d1_decoder_layer.sv	180 total	180 code
rtl/anvil_d1_exp.sv	113 total	113 code
rtl/anvil_d1_expert.sv	291 total	291 code
rtl/anvil_d1_isqrt.sv	57 total	57 code
rtl/anvil_d1_mem_mux.sv	51 total	51 code
rtl/anvil_d1_mem_port.sv	67 total	67 code
rtl/anvil_d1_mla_engine.sv	492 total	492 code
rtl/anvil_d1_moe_router_dma.sv	478 total	478 code
rtl/anvil_d1_mul32.sv	55 total	55 code

Production file	Physical lines	Code lines
rtl/anvil_d1_mul96x32.sv	77 total	77 code
rtl/anvil_d1_parameters.sv	48 total	48 code
rtl/anvil_d1_rmsnorm.sv	207 total	207 code
rtl/anvil_d1_rope.sv	154 total	154 code
rtl/anvil_d1_round_sat.sv	46 total	46 code
rtl/anvil_d1_sdiv.sv	72 total	72 code
rtl/anvil_d1_silu.sv	97 total	97 code
rtl/anvil_d1_sub_unsigned.sv	44 total	44 code
rtl/anvil_d1_tensor.sv	196 total	196 code
rtl/anvil_d1_token_io.sv	130 total	130 code
rtl/anvil_d1_top.sv	331 total	331 code
rtl/anvil_d1_udiv.sv	68 total	68 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E028	rtl/anvil_d1_top.sv
E032	tests/defects.py
E037	tests/floating.py
E040	tests/legacy.py
E043	tests/reference.py
E045	tests/router.py
E049	tests/wide_mla.py
E071	.vulcan/engineering/run-0001/attempt-0016/stdout.log
E072	.vulcan/engineering/run-0001/attempt-0017/stdout.log
E073	.vulcan/engineering/run-0001/attempt-0018/stdout.log
E074	.vulcan/engineering/run-0001/attempt-0019/stdout.log
E075	.vulcan/engineering/run-0001/attempt-0020/stdout.log
E1007	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1008	.vulcan/engineering/run-0001/quartus-sta.json
E1009	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).